

Internship Report: AWS in a Copper Riveting Factory Management System

Introduction

During today's internship session, I learned about Amazon Web Services (AWS) and how cloud computing is used to build real-world applications. Before this session, I had basic knowledge of AWS services, but I wanted to understand how they work together in a complete project instead of learning each service separately.

For this activity, I chose a **Copper Riveting Factory Management System** because manufacturing industries deal with many daily operations such as production planning, inventory management, machine maintenance, employee records, and customer orders. Managing all this information manually or using local servers can become difficult as the factory grows. By using AWS, these operations can be managed more efficiently through cloud technology. This example helped me understand how cloud services can solve practical problems in the manufacturing industry.

Why Cloud Computing?

A factory that runs its management system on its own servers has to purchase hardware, maintain storage devices, install software updates, perform backups, and ensure data security. As production increases, the amount of data also grows, making the system more expensive to maintain.

Cloud computing solves many of these challenges. With AWS, the factory does not need to manage physical infrastructure because AWS provides the required computing resources whenever they are needed. Resources can easily be increased during busy production periods and reduced when demand is lower. This makes the system more flexible, reliable, and cost-effective.

Amazon EC2

Amazon EC2 can be used to host the main factory management application. Through this application, managers can monitor daily production, assign work orders, manage employees, track machine performance, and generate reports.

If the factory receives a large number of customer orders or experiences higher production during certain seasons, additional EC2 instances can be launched automatically to handle the extra workload. This ensures that the application continues to perform smoothly without slowing down.

Amazon S3

A copper riveting factory generates many important files every day, such as production reports, machine maintenance records, inspection photographs, invoices, purchase orders, and delivery documents.

Amazon S3 provides secure and reliable cloud storage for these files. Instead of storing everything on the application server, documents can be uploaded to S3, making storage management easier and improving system performance. Older files that are not accessed frequently can also be moved to lower-cost storage, helping reduce overall storage expenses.

Database Services

The factory needs to store information such as employee details, supplier information, raw material inventory, copper rivet stock, customer orders, production schedules, quality inspection reports, and sales records.

Amazon RDS is suitable for storing this structured data because it automatically manages backups, software updates, and database maintenance. If the application requires very fast access to continuously changing information, such as live inventory levels or machine status, Amazon DynamoDB can also be used.

AWS Lambda

Not every task in a factory requires a person to perform it manually. Some repetitive tasks can be automated using AWS Lambda.

For example, once production is completed, Lambda can automatically update inventory records. It can also generate daily production reports, send maintenance reminders when machines require servicing, notify managers when raw material stock becomes low, generate invoices after customer orders are completed, and send email notifications to customers or staff. Automating these tasks saves time, reduces manual effort, and improves overall efficiency.

IAM and Security

Security is one of the most important parts of any management system. AWS Identity and Access Management (IAM) allows different users to access only the information they need.

For example, factory workers can update only their assigned production tasks. Inventory staff can manage stock records, quality inspectors can upload inspection reports, and maintenance engineers can access machine maintenance details. Factory administrators

have full control over the system, including user management and report generation. This role-based access helps protect sensitive company data and prevents unauthorized access.

Load Balancer and Reliability

At busy times, many employees may use the factory management system simultaneously. An Elastic Load Balancer distributes incoming requests across multiple EC2 instances so that no single server becomes overloaded.

AWS also provides monitoring and backup services that improve system reliability. Even if one server experiences a problem, the application can continue running with minimal interruption, ensuring that important production data remains safe.

Monitoring and Analytics

AWS CloudWatch helps monitor the health and performance of the application. It collects system logs, CPU usage, memory utilization, storage information, and other important metrics.

Managers can use this information to monitor production efficiency, machine performance, inventory levels, and overall factory operations. By analyzing this data, they can identify problems early and make better business decisions.

Conclusion

Today's internship activity helped me understand how different AWS services work together to build a complete cloud-based application instead of functioning as separate services. By using the example of a **Copper Riveting Factory Management System**, I was able to relate each AWS service to a real industrial requirement.

I learned that Amazon EC2 can host applications, Amazon S3 can securely store factory documents, Amazon RDS can manage business data, AWS Lambda can automate repetitive tasks, IAM can provide secure access, and Elastic Load Balancer and CloudWatch help maintain system performance and reliability.

Overall, this activity improved my understanding of cloud computing and showed me how AWS can be used to build secure, scalable, and efficient solutions for modern manufacturing industries. It also gave me a better understanding of how cloud technology can simplify factory operations and support future business growth.